Review Journal of Autism and Developmental Disorders
https://doi.org/10.1007/s40489-020-00223-w

REVIEW PAPER

Caregiver Involvement in Communication Skills for Individuals with ASD and IDD: a Meta-analytic Review of Single-Case Research on the English, Chinese, and Japanese Literature

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Received: 28 February 2020 / Accepted: 1 October 2020
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Abstract

The current meta-analytic review analyzed 43 studies published in English, Chinese, and Japanese to determine the effects of caregiver involvement for promoting communication skills of individuals with ASD and IDD. Tau-U effect sizes, the Kruskal-Wallis H test, and the Dunn post hoc test were employed to calculate for moderator analyses: child age, settings, service delivery formats, and dosages of services provided to caregivers. The overall effect size for family involvement had a moderate effect on child communicative outcomes, as well as on child ages, settings, delivery formats, and dosages of services provided to caregivers of individuals with ASD and IDD. A statistically significant difference was found in children's communication outcomes between the four dosage groups of services provided to caregivers.

Keywords Caregiver involvement · Service delivery · Communication intervention · Autism spectrum disorder · Intellectual/developmental disability

Autism spectrum disorder (ASD) is characterized by impairments in social interaction, language, and communication as well as restricted interests and repetitive behaviors. Intellectual/developmental disability (IDD; or intellectual disability) is characterized by intellectual and adaptive functioning deficits in conceptual, social, and practical domains (DSM-V; American Psychiatric Association 2013). One of the common defining features of ASD and IDD is a functional limitation in communication and social participation like initiations, joint attention, capacity for vocal communication, and symbol use (American Psychiatric Association 2013; Greenspan and Wieder 1999; National Research Council 2001; Sappok et al. 2017). Children with ASD and IDD often fail to develop the ability to initiate conventional communication and establish and maintain social reciprocity for daily needs across different contexts like the home, school, and

Portions of this study were previously published as part of the first author's doctoral dissertation.

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Published online: 26 October 2020

community (Dogoe et al. 2010). Communication abilities such as making requests are important for life outcomes. Interventions that are designed for improving communication skills are important for children with ASD and IDD when deficits in communication are present (Goldstein 2002; Warren 2000). There is an increase in children with ASD and IDD who start to receive interventions before 3 years old, so more and more caregivers have participated in the service delivery system to collaborate with professionals for years before their children receive education at school (Garbacz et al. 2016; Matson et al. 2009).

The deficits in communication and social interaction are generally evident in individuals with ASD before 3 years old (Individuals with Disabilities Education Act, IDEA 2004). Children with ASD and IDD frequently do not communicate at their current age level. However, verbal and non-verbal communication skills in children with ASD and IDD improve when they receive training and treatment from professionals (e.g., teachers and therapists; Ando and Yoshimura 1979; Vicker 2009) or from caregivers at home and community settings (Chaabane et al. 2009; Gerow et al. 2018; Koegel et al. 2002; Lee 2015a; Zimmer 2013). Behavioral interventions are most commonly provided to young children with ASD and IDD; thus, the majority of previous research has aimed to investigate communication outcomes of young children with ASD and IDD (Hong et al. 2016a).

Caregiver Involvement

Collaborating with caregivers and family in the intervention is very important for children with ASD and IDD (Mackintosh et al. 2012; Tzanakaki et al. 2012). In natural environments, it is critical that caregivers provide more opportunities for communication, a richer environment, and great life experience for children with ASD and IDD (De Houwer 1999). Caregiver involvement is an individual-centered connection between caregivers at home and professionals in school settings (e.g., teachers, therapists, and service providers) who share responsibility for the development of individuals with ASD and IDD (Garbacz et al. 2016; Johnson et al. 2018). The goal of caregiver involvement is to provide knowledge, parenting skills, and support to caregivers for promoting and maintaining the improvements of communication and behaviors in children with ASD and IDD (Johnson et al. 2018; Matson et al. 2009). Six types of parent involvement include (a) providing parenting skills and knowledge to caregivers, (b) communicating with caregivers, (c) involving families as volunteers, (d) involving caregivers and children learning at home, (e) including caregivers as participants in educational decision making, and (f) collaborating with the community (Epstein and Salinas 2004; Epstein et al. 2002). Common types of caregiver involvement for caregivers of individuals with ASD and IDD include caregiver-implemented interventions, caregiver training, education, and coaching in which caregivers learn from professionals in an attempt to put the strategies into practice when they work with their children (Brown and Woods 2015; Powers et al. 1992; Robertson 2016; Schreibman et al. 2015; Wacker et al. 2005).

Caregiver-implemented interventions in natural environments are highly flexible, easily generalized, and low cost (Brown and Woods 2015; Powers et al. 1992; Robertson 2016; Schreibman et al. 2015; Wacker et al. 2005). After receiving training from professionals, caregivers with limited experience are able to implement behaviorally based strategies (e.g., modeling and prompting), communication interventions (e.g., AAC and PECS), and use more language facilitative practices. Children receiving caregiver-implemented intervention also show a significant increase in social utterances and a decrease in inappropriate communication behaviors (e.g., repetitive speech) in natural contexts (Becker-Cottrill et al. 2003; Binger et al. 2008; Bruno and Dribbon 1998; Goldstein 2002; Hong et al. 2016a, 2016b; Hsieh et al. 2011; Johnson et al. 2007; Matson et al. 2009; Patterson and Smith 2011; Powers et al. 1992; Tekin-Iftar 2008). With caregiver involvement, children with ASD and IDD gain more opportunities to practice communication and generalize learned skills to different settings, activities, and with different communication partners (Dogan et al. 2017; Douglas et al. 2018; Hansen et al. 2018; Liu et al. 2015; Wright and Kaiser 2017). However, most reviews and studies regarding family involvement with children with ASD and IDD have focused on how children's backgrounds (e.g., age, diagnosis, severity, and communication mode), components of interventions (e.g., settings, education delivery, and duration of interventions) provided to children, and parent stress affect outcomes in children with ASD and IDD (Hayes and Watson 2013; Nevill et al. 2018; Strauss et al. 2013; Preece and Trajkovski 2017). There is a need to conduct more studies and reviews exploring effective methods of service provision to caregivers of children with ASD and IDD.

Compared with other countries in Asia, an increased prevalence of ASD was reported in Japan and China (Sun and Allison 2010). In China, the prevalence of ASD was 0.32 per 10,000 population in 1987, 9.8 per 10,000 population in 2008, and increased to 39.23 per 10,000 population in 2018. The prevalence of ASD in Japan was 2.6 per 10,000 population in 1971, 13 per 10,000 population in 2008, and 322 per 10,000 population in 2016 (Saito et al. 2020; Sun and Allison 2010; Wang et al. 2018). Asian cultural beliefs tend to have stereotypes and negative images about those with disabilities, which affect service provision to families of children with ASD and IDD (Chao et al. 2018; Kayama 2010). In China, the lack of awareness and knowledge about ASD in public and professionals restricts the development of the healthcare system and special services for individuals with ASD, which aggravate the burden on families of children with ASD (Sun et al. 2012; Zhang et al. 2019). In Taiwan, although there is an

increase of awareness and knowledge of ASD and IDD, parents of children with ASD and IDD still experience high levels of stress and anxiety. Children with ASD and IDD are able to receive special education and services based on parents' decisions or doctors' referrals; however, short visits with the doctor take a toll on the diagnosis and parents' understanding of their child's special needs. For example, some doctors spend only an average of 3–5 min with each patient. Parents of children with ASD and IDD do not have sufficient time to discuss their child's special needs with the doctor. Also, the delayed services provided by formal social support network (which is usually composed of a leading physician, social worker, special educator, developmental psychologist, and therapist) was mentioned by parents of children with ASD and IDD (Chao et al. 2018; Sung 2008). In Japan, educating children with disabilities is considered a responsibility of parents rather than social services. However, when talking about a child's disability and requesting services and support, Japanese parents usually feel guilt and shame due to the lack of communication and information sharing from professionals, low awareness of professional's awareness of disability, and adverse effects on competition among professionals. Japanese parents are eager to establish empathetic and trusting relationships when collaborating with professionals (Kayama 2010). Also, in Asia, caregiver involvement in the implementation of interventions for communication outcomes of children with IDD (e.g., parent training, parent-implemented intervention) is a new field of study in recent decades (Guan 2002). Studies conducted in Asia suggest that family involvement in interventions decreases in challenge behaviors like screaming, as well as increases in life skills and verbal and nonverbal communication skills (e.g., joint attention, social orienting, and AAC use). In addition, caregivers' knowledge of ASD interventions was increased, and they were able to implement and generalize learned strategies and interventions (e.g., environmental arrangement, reinforcement delivery, and pivotal response treatment) across different settings (Chu et al. 2015; Hagimori 2004; Kuramitsu et al. 2008; Lee 2015b; Liou et al. 2005; Lu, 2011; Wang 2007; Yang and Ge 2015; Zhang et al. 2019).

Although these studies concerning caregiver involvement in children with ASD and IDD have been published in Chinese and Japanese, literature-based discoveries regarding family involvement in individuals with ASD and IDD remain poorly understood in Asia. A dearth of studies and reviews across cultures have explored the components of services (e.g., settings of delivering services, delivery formats of services provided to caregivers, and dosage of services) that are provided to caregivers of individuals with ASD and IDD. Thus, there is an urgent need to probe into how these components of service delivery influence caregiver implementation and child communication outcomes.

Single-Case Research and Effect Size

Single-case research is a methodology of scientific research widely applied in the field of special education to evaluate the effectiveness of educational practice (Horner et al. 2005; Kratochwill et al. 2013). Visual analysis is one method used to interpret data from single-case experimental studies. There are 6 elements for evaluating causality in single-case studies, including level, variability, trend in the data path, overlap of data points between phases, immediacy of effects, and consistency across similar phases. When using these elements, researchers should check data within and between phases, compare data across phases, and then integrate the information from all phases (Kennedy 2005; Vannest 2016). Recent studies have increased the objectivity of single-case research by conducting data analysis, visual analysis, and effect sizes altogether. Tau-U is a quantitative approach that combines non-overlap between phases with the trend of intervention phases and controls a baseline trend (Parker, Vannest, Davis, and Sauber 2011b). In meta-analysis, effect sizes can be used to compare intervention effects across studies and moderators. Since previous single-case studies were mostly conducted to evaluate the effects of caregiver involvement in children with ASD and IDD, the information about the components of caregiver involvement and service delivered to caregivers can be extracted from these studies. Thus, a comprehensive review should be conducted to obtain a better understanding of the support and services provided to caregivers of individuals with ASD and IDD for family involvement.

Purpose of the Current Study

A meta-analysis was conducted to determine the effects of caregiver involvement for promoting communication skills of children with ASD and IDD. The purpose of the current meta-analysis is to address the following:

1 What are the overall effects of caregiver involvement for promoting communication skills of children with ASD and IDD?
2 Do these effects differ by child age?
3 Do these effects differ by settings of delivering services to caregivers?
4 Do these effects differ by delivery formats of services provided to caregivers?
5 Do these effects differ by different dosages of services provided to caregivers?

Method

To achieve the goal of the current meta-analysis, the following procedures (i.e., article identification, data extraction, and

inter-rater reliability) based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram (Moher, Liberati, Tetzlaff, Altman, and the PRISMA Group, 2009) were conducted from November 2018 (see Liao et al. 2019a for the flow chart depicting the search process for selecting eligible studies).

Article Identification

Search Strategy

This review included studies in English, Chinese, and Japanese. The groups of search terms combined to search within each database were as follows: (a) parent-mediated, parent-based, parent-implemented, parent-directed, parent-involved, caregiver-mediated, caregiver-implemented, caregiver-involve, family-mediated, family-based, family-implemented, family-involved; (b) communicat*, social*, language; and (c) autism spectrum disorder, asd, autism, autistic symptoms, autis*, Asperger*, pervasive developmental disorder not otherwise specified, PDD-NOS, childhood disintegrative disorder, CDD, Rett, intellectual dis*, and developmental dis*. Also, all English search terms were reviewed and translated separately to Chinese and Japanese by a native Chinese speaker and a Japanese speaker to confirm the accuracy of translation (see Liao et al. 2019b). Harvested synonyms for each group were combined with OR, and all three concept lists were combined using AND.

Eight English electronic databases were searched for articles and dissertations, including Academic Search Complete, Education Resources Information Center (ERIC), Education Full Text, Medline, Professional Development Collection, PsycINFO, Social Sciences Full Text, and Proquest Dissertations & Theses Global. Also, to include studies conducted across different countries, 6 additional Chinese databases were searched for journal articles, grant reports, and dissertations in Chinese, including National Applied Research Laboratories' Patent and Research Paper platform Search System, PPSEARCH (國家實驗研究院-專利與學術文獻檢索系統), Government Research Bulletin (政府研究資訊系統), PerioPath Index to Taiwan Periodical Literature System (臺灣期刊論文索引), National Digital Library of Theses and Dissertations in Taiwan (臺灣博碩士論文知識加值系統), China Academic Journals Full-text Database (中国期刊全文数据库), and China Doctoral Dissertations Full-text Database (中国博士学位论文全文数据库). Also, 2 Japanese databases (i.e., CiNii-NII学術情報ナビゲータ, サイニィ and CiNii Dissertations - 日本の博士論文) were searched for journal articles and dissertations in Japanese. The combinations of search terms varied due to the limitation of each search system (e.g., length of search terms). The initial search obtained a total of 4713 records. Specifically, there were 2076 records in English, 2295 records in Chinese, and 342 in Japanese. Then, after identifying records through searching the electronic databases and removing duplicates, all titles and abstracts of the articles were screened for further evaluation.

Title and Abstract Review

For the title and abstract review, duplicates were removed among these records. Rayyan is a free web application for systematic reviews and meta-analysis. Due to the limitation of Chinese and Japanese databases, records from English databases were exported to Rayyan, and records from Chinese and Japanese databases were exported to an Excel document for evaluation. The title or abstract inclusion criteria included the following: (a) at least one participant was diagnosed with ASD (e.g., autism, ASD, autistic symptoms, Asperger syndrome, pervasive developmental disorder not otherwise specified (PDD-NOS), Rett's Disorder, or childhood disintegrative disorder (CDD), and Kanner's Syndrome) or IDD (e.g., intellectual and developmental disabilities, global developmental disability, Down syndrome, and cerebral palsy), which were redefined in the DSM-V and other developmental disabilities in the DSM-IV (American Psychiatric Association 2013). (b) The study included an intervention about family involvement in which caregivers (e.g., parent, caregiver, family, father, mother, and grandparent) received training, coaching, and education to implement intervention (e.g., parent training, parent coaching, parent-implemented, parent-directed, and parent-mediated). (c) At least one dependent variable was relevant to child communication skills. (d) The research design was a single-case experimental study. (e) The document was written in English, Chinese, or Japanese (Liao et al. 2019c). When the title and abstract review were not able determine if the record met the inclusion criteria, the full-text review of that document was conducted.

Full-Text Review

The full text of the records was reviewed to identify the included records after the title and abstract review. Several inclusion criteria were used to review and screen the full text of each article. First, the article was written in English, Chinese, or Japanese. Second, the research design in each article was a single-case experimental design with at least one of the time series graphs that included AB contrasts about communication outcomes in children with ASD and IDD. Third, at least one child participant in each study was diagnosed with ASD or IDD. Fourth, the independent variable had to include family involvement in which parents, caregivers, and families collaborated with professionals to work with their children with ASD and IDD. The interventions that parents learned during the intervention phase must be evidence-based practices. Fifth, the dependent measures were related to child communication skills. Communication skills were defined as skills

related to expressive language (e.g., initiations, response, or commenting), linguistic changes (e.g., target morpheme, mean length of the utterances, or language samples), or social play (Liao et al. 2019c). After reviewing the full text of each study and excluding 5 duplicate studies, 46 records were included for methodological quality.

Methodological Quality Standard Review

The methodological quality standard reviews were conducted to evaluate if these 46 included studies met the methodological quality standards. A set of methodological quality criteria for the single-case experimental study proposed by previous studies was utilized to develop the current quality standards (Council for Exceptional Children 2014; Ganz and Ayres 2018; Horner et al. 2005, Kratochwill et al. 2013, Maggin et al. 2013; Reichow et al. 2008; U.S. Department of Education, Institute of Education Sciences, What Works Clearinghouse 2016). This quality standard included three criteria. The first criterion is basic methodological standards to evaluate each included study in independent variable, inter-observer agreement, experimental control, and number of data points per phase. The second criterion is extended methodological standards to evaluate description of participant, setting, material, interventionist, dependent variable, baseline and intervention, maintenance data, generalization data, procedural integrity, and social validity of each included studies. Then, for multiple probe design, additional standards were used to evaluate quality. Five duplicate studies were excluded. After methodological quality standard review, 26 studies met the basic methodological standards for reservations, and 15 studies did not meet the basic methodological quality standards.

First Author, Ancestral, and Forward Review

To obtain more comprehensive information from studies regarding family involvement for individuals with ASD and IDD across different cultures, a first author search and ancestral search were conducted with all 41 records. Forward searches were conducted only on 26 studies that met the basic standards or met them with reservations, indicating only studies that met basic methodological quality standards were cited from the search articles. First author, ancestral, and forward searches were conducted using Scopus after identifying the included articles in methodological quality standard review. Each included record was searched on Scopus for (a) other studies of the first authors of papers that were included after methodological review (i.e., first author search), (b) studies on reference lists of the included articles (i.e., ancestral search), and (c) other articles cited in the included records (i.e., forward search), Then, 1424 records were exported from the search to conduct the title and abstract review, full-text review, and methodological quality review. There were 2 additional articles included in this meta-analysis after first author, ancestral, and forward review.

Moderator Coding

After all procedures of searching and screening for identification were conducted, there were 43 studies included in this meta-analysis. A coding system was developed by the researcher for coding and data analysis. All 43 included articles were reviewed and coded to collect required information about moderator variables for effect size calculation. Each study was reviewed and coded by each experiment for information required to calculate effect sizes and information related to the following four major moderators, including (a) publication characteristics, (b) child age, (c) settings where professionals delivered services to caregivers, (d) delivery formats of services provided to caregivers, and (e) dosages of services provided to caregivers (see Table 1 for the code and description of moderator). The moderator coding produced 170 records.

Data Extraction

Plot Digitizer, a free Java program, was used to extract data points from all *X*-*Y* type scatter or line plots of included articles. Each graph was scanned in JPEG format and saved to an Excel sheet tab organized by each article. Digitized values of data points in baseline and intervention phases of each phase contrast retrieved from Plot Digitizer were exported, organized, and saved in an Excel document for each of these 170 records.

Data Analysis

Tau-U is a non-parametric effect size that combines non-overlap data across different phases that can correct for a baseline trend. Compared to other non-parametric effect sizes, the advantages of Tau-U used in single-case research include the use of all data points, the ability to control for trend, high sensitivity, and ease of calculation (Parker et al., 2011a). The current meta-analysis used Tau-U to calculate an effect size for each AB phase of the included studies. After coding moderator variables, a free online Tau-U calculator (Vannest et al. 2016) was used to calculate the effect size for each baseline-intervention contrast for each participant in each moderator. All effect sizes in one study were combined in the Tau-U calculator to produce one effect size for each included experiment to evaluate which characteristics of participants or interventions improve communication outcomes in individuals with ASD and IDD. Tau-U scores ranged from −1.0 to 1.0, with the positive scores meaning improvement and negative scores meaning deterioration of the data (Parker et al., 2011b).

Table 1 Moderator coding and description

Moderator		Coding	Description
Publication characteristics	First author's last names		First author's last name
	Year of publication		Publication year
	Type of publication	• Journal article • Book or chapter • Dissertation • Technical report • Conference paper • Others	
	Country of publication	• United States • Taiwan • China • Japan • Other countries	
Participant characteristics	Child age	• 1–3 years old • 4–6 years old • > 7 years old	Age of each child participant
Intervention characteristics	Setting	• Home • Clinic/hospital/center • Community settings • More than 2 settings • Not reported	Settings that professionals delivered services to caregivers
	Delivery Format	• In person • Tele-practice/online • Not reported	Types of coach/training/services provided to caregivers
	Dosage	• 1–3 sessions • 4–6 sessions • 7–9 sessions • >10 sessions	How many coach/training/service sessions provided to parents

Benchmarking and interpretation for Tau-U was calculated in a manner consistent with the purpose and expressed guidelines for effect sizes (Cohen 1988). The benchmarks were calculated based on the distribution of a meta-analysis. Contextually, we identified five categorical bands of interpretation, very large, large, typical, small, and very small. Effect sizes were considered very large in the 90th percentile rank, large between the 75th and 89th percentile rank, moderate between the 25th and 75th percentile rank, small in the 10th and 24th percentile rank, and very small below 10th percentile rank (Ganz et al. 2017; Vannest 2019). Stata® (StataCorp 2017) was used to create the visual forest plot.

In addition, the Kruskal-Wallis *H* test was used to evaluate the statistical significance for each potential moderator. The Kruskal-Wallis *H* test is a rank-based non-parametric one-way analysis of variance (ANOVA) to evaluate statistical significance between two or more independent variables (Kruskal and Wallis 1952). The current study conducted the Kruskal-Wallis *H* test for each moderator group to evaluate if there was a significant difference between different groups in a moderator. Also, a Dunn post hoc test was conducted to examine pairwise combinations across sub-groups if a statistical significance was founded (Dunn 1964; Hong et al. 2016a).

Publication Bias

RStudio Team, a modular platform of statistical analysis software, was employed to produce the funnel plot and conduct Egger's regression test for assessing publication bias (RStudio Team 2019). Then, the trim-and-fill method was conducted to correct funnel plot asymmetry due to publication bias and calculate the unbiased effect size (Duval and Tweedie 2000). Publication bias may occur when severe asymmetry in a funnel plot or *p* value less than 0.05 in Egger's regression test (Egger et al. 1997; Stanley and Doucouliagos 2014; Sterne and Harbord 2004; Wattanawongwan 2020).

Inter-rater Reliability

Inter-rater reliability (IRR) was conducted by four trained doctoral students in special education for each procedure during the initial search. A minimum of 50% of the included studies in the title and abstract review and moderator coding, as well as the initial, first author, ancestral, and forward searches were independently coded twice. Also, at least 50% of the included articles were evaluated to ensure the inter-rater reliability of moderator coding and data extraction.

All raters received coding training provided by the author to understand the procedures of coding and the coding system. Before independent coding, each coder coded one study and then discussed the coding results. Retraining was provided if the percentage of agreement was lower than 80%. Each included article was coded by the author, and a second coder independently used the same procedures to code the included articles chosen at random. The results of each article were then compared, and then disagreements were discussed and ultimately reconciled by a third trained coder. Also, to examine IRR for data extraction, IRR was calculated on the agreement of data extraction for each article. The coders discussed the results if there was disagreement on data extraction between two raters. IRR was calculated by dividing the total number of agreements by the total number of agreements plus disagreements and multiplying the sum by 100 to obtain a percentage (Hong et al. 2016a; Morin et al. 2018).

All search terms were used to search each electronic database by the first author and other coders, resulting in 100% agreement of the search results. In the title and abstract review, IRR was collected on 51% of the included records to get 93.93% of agreement. In full-text review, 68.19% of the included records were reviewed by the author and one of the coders, thus obtaining 94.43% of agreement. Then, 65.12% of the included studies were reviewed by the author and one of the coders, and the average percent agreement of methodological quality was 88.67%. Regarding IRR for data extraction, the author and 2 raters independently extracted 39.53% of the included studies, obtaining 93.64% of agreement. For moderator coding, 69.76% of the included studies were coded, obtaining 89.12% of agreement (see Table 2 for articles coded/databases searched and average percent agreement for IRR).

Results

Overall Effects

A total of 43 studies with 170 AB phase contrasts across 149 caregiver-child dyads/triads were included (see Appendices A and B; Liao et al. 2020). After benchmarking for Tau-U values in the included studies, the interpretive guidelines for caregiver involvement for individuals with ASD and IDD were very large when values exceeded 0.995; large effects were Tau-U values ranging from 0.86 to 0.994. Medium Tau-U values were 0.49 to 0.85 and for small effects, 0.21 to 0.48. Very small effects had Tau-U values of 0.20 and below. Effect sizes were separately calculated and then aggregated to obtain omnibus effect sizes for each included study. The overall effect of caregiver involvement on communication skills of children with ASD and IDD was 0.74 with a 95% confidence interval of 0.70 to 0.78 for 170 AB contrasts. Thus, family involvement had a moderate effect of 0.74 on child communicative outcomes (see Fig. 1 and Appendix B; Liao et al. 2020). The effects of individual studies ranged from −0.40 to 1.00. The lowest 20% of effect sizes fell below a Tau-U of 0.34, and the top 20% of studies fell between 0.88 and 1.00 improvement. Forty-one of the 43 included studies were positively skewed with a high number of positive behavior changes. However, one study (Robertson et al. 2013) showed negative behavior changes, and one study (Hsueh et al. 2013) showed no behavior change.

Child Age

To investigate the moderating effect of child age on the effect of caregiver involvement on communication skills of children with ASD and IDD, there were a total of 149 dyads/triads and 170 AB phase contrasts included. Eighty-two child participants were 1–3 years old, 45 children were 4–6 years old, and 22 children were greater than 7 years old. The results showed that child age had a moderate effect of 0.639 CI 95 [0.588, 0.690] on children's communication outcomes. The results showed moderate effects that ranged from 0.690 CI 95 [0.529, 0.851] for those greater than 7 years of age, 0.642 CI 95 [0.551, 0.734]) for ages 4–6 years, and 0.629 CI 95 [0.563, 0.695] for ages 1–3 years. Also, the Kruskal-Wallis H test was conducted to determine if children's communication outcomes were different at different ages. The results showed no statistically significant difference in children's communication outcomes among the three age groups, $\chi^2(2) = 0.920$, $p = 0.631$ (Table 3 and Fig. 2).

Settings

Regarding the third question about different setting(s) of services provided to caregivers, there was a total of 145 dyads/triads and 166 AB phase contrasts. Eighty-seven caregiver participants received services at home, 36 caregivers at clinic or center settings, and 22 caregivers at multiple settings. The results showed that setting(s) had a moderate effect of 0.636 CI95 [0.583, 0.688] on child communication outcomes. The settings resulted in moderate effects for caregiver involvement in prompting communication on individuals with ASD and IDD (ES = 0.669 CI95 [0.603, 0.734] for home setting; ES = 0.583 CI95 [0.444, 0.721] for multiple settings; and ES = 0.572 CI95 [0.459, 0.685] for clinic/hospital/center setting; Fig. 3). Also, the Kruskal-Wallis H test was conducted to determine if children's communication outcomes were different at different settings. The results showed no statistically significant difference in children's communication outcomes among the three groups, $\chi^2(2) = 4.494$, $p = 0.106$ (Table 3).

Table 2 Articles coded/databases searched and average percent agreement for inter-rater reliability

	Percentage of databases searched/percentage of articles coded	Percentage of agreement
Initial search in databases	100% of databases	100%
Title and abstract review	51%	93.93%
Full-text review	68.19%	94.43%
Methodological quality review	65.12%	88.67%
First author, ancestral, and forward search	100%	100%
Data extraction	39.53%	93.64%
Moderator coding	69.76%	89.12%

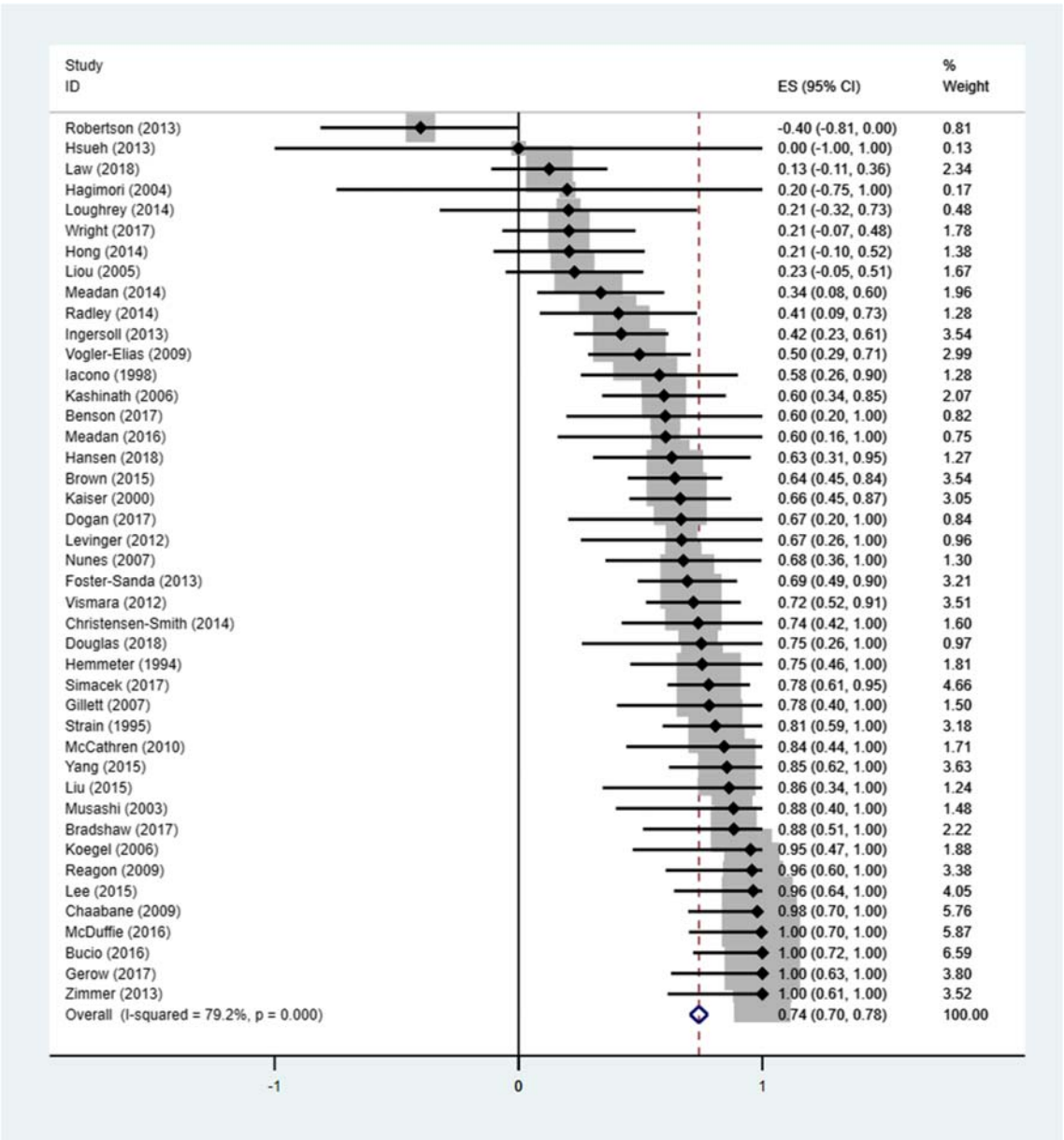


Fig. 1 Forest plots and effect size data for included studies

Table 3 Results of Tau-U effect sizes and Kruskal-Wallis test for the moderators

Moderator	Sub-groups	Number of studies	AB phase contrasts	Number of participants (dyads/triads)	Number of pairwise comparison	Tau (95%CI)	*p* values	Kruskal-Wallis
Age						0.6390 (0.5880, 0.6900)		$\chi^2(2) = 0.920$, $p = 0.6312$
	1–3 years	28	89	82	10,281	0.6287 (0.5625, 0.6950)	0.000	
	4–6 years	21	54	45	5676	0.6424 (0.5507, 0.7340)	0.000	
	> 7years	11	27	22	2423	0.6900 (0.5288, 0.8511)	0.000	
Setting						0.6360 (0.5830, 0.6880)		$\chi^2(2) = 4.494$, $p = 0.1057$
	Home	25	104	87	12,247	0.6686 (0.6032, 0.7340)	0.000	
	Clinic/hospital/center	8	36	36	3341	0.5721 (0.4592, 0.6850)	0.000	
	Multiple settings	8	26	22	2542	0.5825 (0.4438, 0.7212)	0.000	
Delivery formats						0.6280 (0.5760, 0.6810)		$\chi^2(1) = 1.284$, $p = 0.2572$
	In person	36	135	120	14,504	0.6017 (0.5422, 0.6613)	0.000	
	Tele-practice	7	34	28	3807	0.7156 (0.6081, 0.8231)	0.000	
Dosage						0.6250 (0.5740, 0.6760)		$\chi^2(3) = 17.239$, $p = 0.0006$
	1–3 sessions	5	15	13	1864	0.7170 (0.5065, 0.9275)	0.000	
	4–6 sessions	13	24	23	1856	0.7361 (0.5964, 0.8758)	0.000	
	7–9 sessions	8	29	22	2822	0.7574 (0.6380, 0.8768)	0.000	
	> 10 sessions	24	87	84	11,209	0.5530 (0.4882, 0.6179)	0.000	

Service Delivery Formats

The effect of different delivery formats of services provided to caregivers on caregiver involvement for communication skills of children with ASD and IDD demonstrated a statistically significant difference. A total of 148 dyads/triads and 169 AB phase contrasts were included among the 43 studies. One hundred and twenty caregiver participants received services from professionals in person, and 28 caregivers received services via tele-practice. The results of delivering services to caregivers had a moderate effect of 0.628 CI95 [0.576, 0.681] on child communication outcomes (see Fig. 4). The delivery

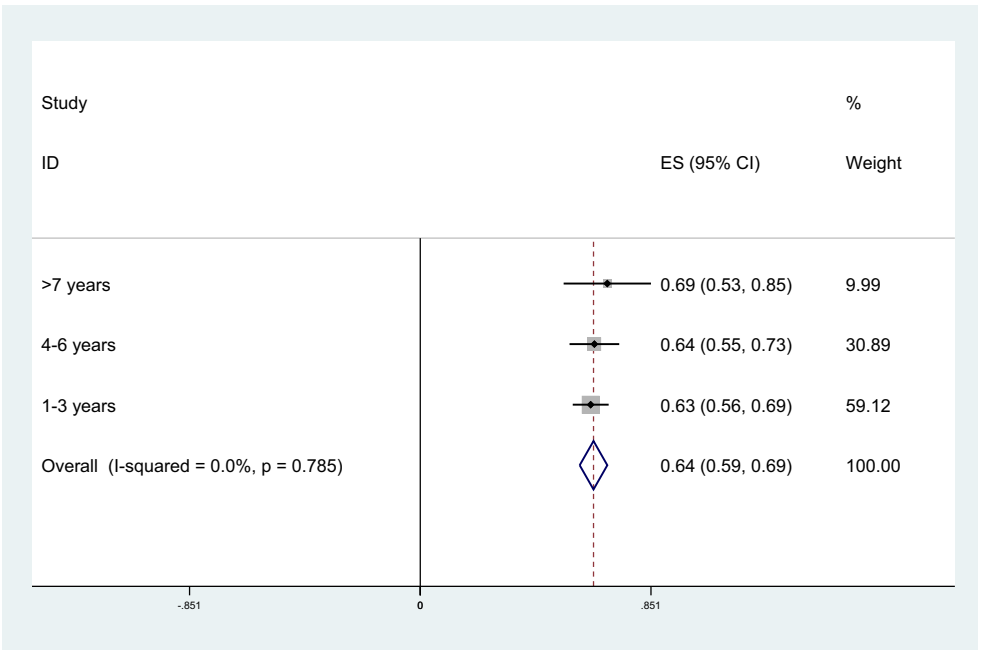


Fig. 2 Forest plots and effect size data for moderators in child age

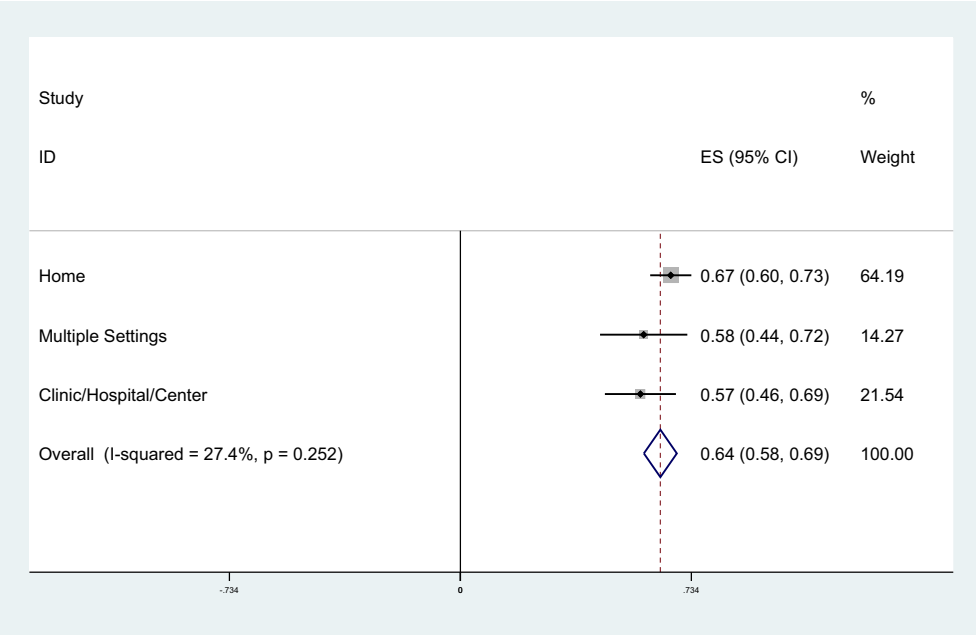


Fig. 3 Forest plots and effect size data for moderators in settings

formats of services resulted in moderate effects for delivering services via tele-practice (ES = 0.716 CI95 [0.608, 0.823]) and for delivering services in person (ES = 0.602 CI95 [0.542, 0.661]). Also, the Kruskal-Wallis *H* test was conducted to determine if children's communication outcomes were different at different delivery formats. The results showed no statistically significant difference in children's communication outcomes between the two groups, $\chi^2(1) = 1.284$, $p = 0.257$ (Table 3).

Dosages

As for the fifth question concerning the effects of different dosages of services provided to caregivers on caregiver involvement for communication skills of children with ASD and IDD, a total of 142 dyads/triads and 155 AB phase contrasts were included. Thirteen caregiver participants received services from professionals for 1–3 sessions, 23 parents for 4–6 sessions, 22 parents for 7–9 sessions, and 84 parents for more than 10 sessions. The dosage of delivering services to caregivers had a moderate effect of 0.625 CI95 [0.574, 0.676] on child communication outcomes (see Fig. 5). Tau effect sizes ranged from 0.757 CI95 [0.638, 0.877] for 7–9 sessions, 0.736 CI95 [0.596, 0.876] for 4–6 sessions, 0.717 CI95 [0.507, 0.928] for 1–3 sessions, and 0.553 CI95 [0.488, 0.617] for more than 10 sessions.

The Kruskal-Wallis *H* test was conducted to determine if children's communication outcomes were different at four dosage groups: (a) 1–3 sessions ($n = 15$), (b) 4–6 sessions ($n = 24$), (c) 7–9 sessions ($n = 29$), and (d) more than 10 sessions ($n = 87$). The results showed a statistically significant difference in children's communication outcomes among the four groups, $\chi^2(3) = 17.239$, $p = 0.0006$ (Table 3). Results of the Dunn's post hoc test revealed that caregivers receiving more than 10 sessions were statistically significantly different from the other three dosage groups in child communication outcomes ($p = 0.0033$ from the 1–3 session group, $p = 0.0043$ from the 4–6 session group, and $p = 0.0009$ from the 7–9 session group).

Publication Bias

The funnel plot and Egger's regression test were conducted for publication bias. Although the funnel plot asymmetry (see Fig. 6 for the funnel plot) suggested potential publication bias, the result of Egger's regression test showed the absence of publication bias ($z = -0.403$, $p = 0.687$). Further, since a statistically significant result of the trim-and-fill method ($Q = 133.799$, df = 42, $p < 0.0001$) indicating the heterogeneity of the effect sizes, the adjusted mean effect size was calculated and resulted in 0.643 CI95 [0.555, 0.730] with 0.044 variability.

Discussion

This meta-analytic review investigated single-case research published in English, Chinese, and Japanese on caregiver involvement in communication skills for individuals with ASD and IDD. After analyzing 43 studies in the form of single-case experimental designs with 170 AB phase contrasts across 149 caregiver-child dyads/triads, the overall effect size for family

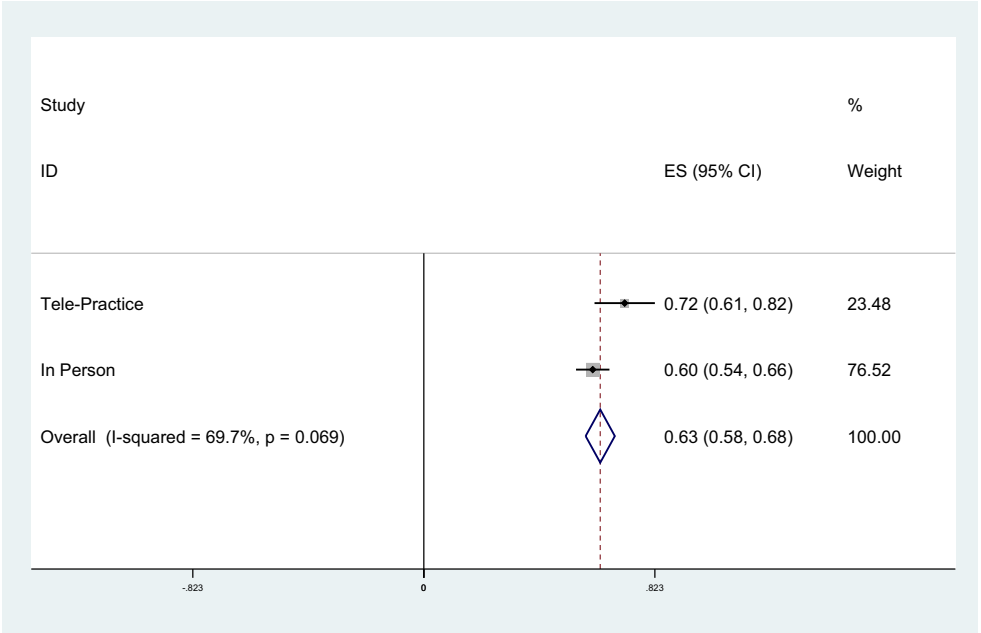


Fig. 4 Forest plots and effect size data for moderators in delivery formats of services

involvement had a moderate effect on child communicative outcomes, as well as on child ages, settings, delivery formats, and dosages of services provided in caregiver involvement in communication skills of children with ASD and IDD. A statistically significant difference in children's communication outcomes between the four dosage groups was also found.

Regarding child age, children's age showed a moderate effect on their communication behaviors. The effect size of the adult group was higher than that of the child and preschool groups, which corresponds to the research finding that the receptive and expressive language of children with ASD significantly improved with age (Ando and Yoshimura 1979). Older individuals with ASD and IDD have better skills in comprehension and conversation. However, very few studies ($n = 3$) included adolescents and adults with ASD and IDD. Empirical studies about family involvement in improving communication skills of individuals with ASD and IDD are lacking. The results of this meta-analytic review correspond to the findings of related studies indicating that caregivers are able to correctly implement interventions for the improvement of children's communication skills in natural contexts (Bruno and Dribbon 1998; Goldstein 2002; Hong

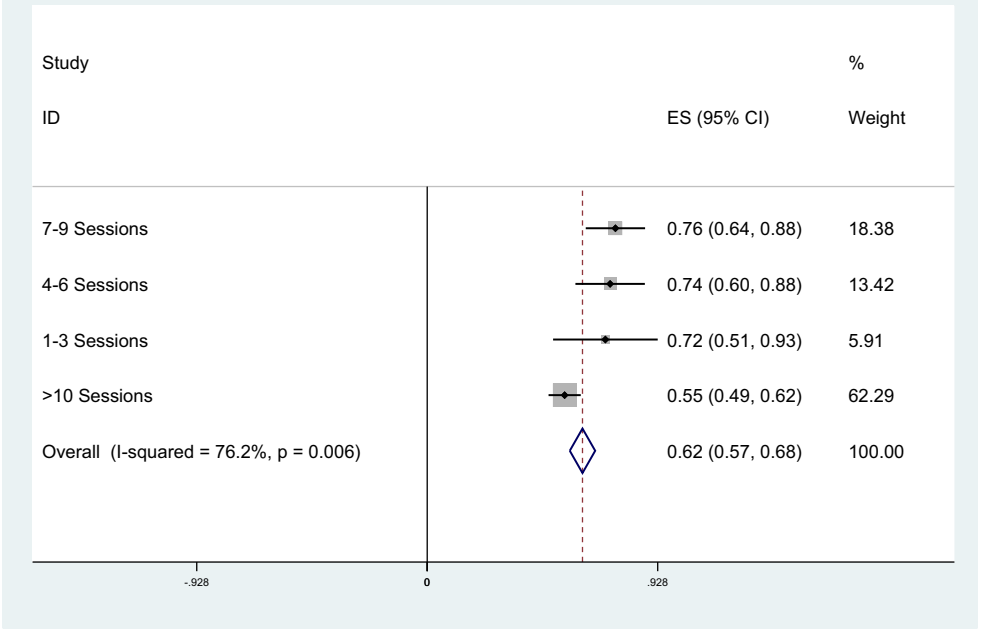


Fig. 5 Forest plots and effect size data for moderators in dosage

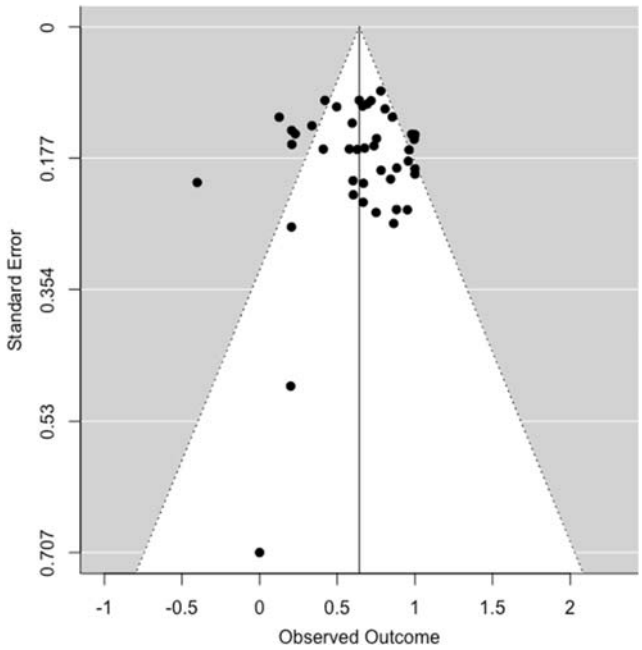


Fig. 6 Funnel plot of included studies for assessing publication bias

et al. 2016a; Patterson and Smith 2011; Powers et al. 1992; Tekin-Iftar 2008). However, the effect sizes across three age groups (i.e., 1–3 years, 4–6 years, and more than 7 years) showed that children's ages had low effects on child communicative outcomes. Only 9 adolescents and adults with ASD and IDD greater than 7 years of age were included in this meta-analysis, which correspond to a related finding that most studies investigated communication outcomes of young children with ASD and IDD (Hong et al. 2016a).

Caregiver involvement has shown reductions in caregivers' stress and anxiety and improvement in caregiver-child interaction, parental quality of life, the understanding of ASD and IDD, and communication skills of individuals with ASD and IDD. This meta-analytic review expanded the existing reviews and the studies about caregiver implementation and involvement to improve communication outcomes of children with ASD and IDD. The results showed that settings, dosage, and formats of delivering services (e.g., parent coaching, parent education, and parent training) had low effects on communication outcomes of individuals with ASD and IDD. Furthermore, differences of child communication outcomes among different dosages were statistically significant. More dosage (e.g., more parent coaching sessions) provided to caregivers did not indicate more improvements in communication outcomes of individuals with ASD and IDD. Some caregivers in the included studies did not complete all training sessions. The reasons for high dropout rates from parent training include low satisfaction, low socioeconomic status, the contents of parent training, and the achievement of treatment goals (Forehand et al. 1983; Matson et al. 2009). It is not necessary for caregivers to receive training for more than 10 sessions to correctly implement intervention strategies with their children with ASD and IDD. Instead of a specific number of sessions provided to caregivers, setting fidelity criteria may be another option of evaluating the amount of dosage provided to caregivers.

Although caregivers' characteristics were not included as a moderator, the current study found that most of the included studies reported very limited information regarding caregiver characteristics. Most experimental studies and reviews focused more on child characteristics and outcomes. Thus, it is hard to evaluate how caregiver characteristics affect child communication performance of individuals with ASD and IDD. However, understanding the backgrounds of caregivers is important for practitioners to provide more appropriate services and support to meet the unique needs of each family (Patterson et al. 2012). Experimental studies providing more information about participant characteristics would be helpful in explaining research results and for the benefit of policy and practice.

There are some limitations to the current review and suggestions, which await future investigation. First, this review included only single-case research. However, some studies with the use of group design and mixed methods research were found during the search procedures. Future meta-analyses might include different research designs to evaluate the comprehensive effects of caregiver involvement. Second, this review focused more on intervention characteristics provided to caregivers. Future research might include the characteristics of caregivers (e.g., age, educational backgrounds, and socioeconomic status) to understand how caregiver's backgrounds affect the outcomes of the implementation of intervention and child communication outcomes. Third, although this review attempted to include studies published in Asia, very few articles were included during the screening process. Thus, it was hard to evaluate how cultural differences affect family involvement. Future researchers might use different methods (e.g., systematic review) or revise inclusion and exclusion criteria to include more studies across different cultures. Subsequently, a better understanding of caregiver involvement for the improvement of communication outcomes of children with ASD and IDD across different cultural contexts could be found. Lastly, based on the results of this meta-analysis, the included studies did not provide details about the characteristics of service delivery for caregivers of individuals with ASD and IDD (e.g., the types of services provided to caregivers). Since delivery, content, and contexts may impact parent education for those with a child who has ASD (Preece and Trajkovski 2017), future single-case research may provide more details of service delivery to caregivers for developing the most appropriate protocol about service provision to caregivers who have children with ASD and IDD. Also, few adolescents and adults with ASD and IDD participated in the included single-case studies. There is an inevitable need to

conduct more single-case studies to evaluate the effects on caregiver involvement in communication outcomes of adolescents and adults with ASD and IDD.

Compliance with Ethical Standards

Conflict of Interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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